

Project Report on

BlogSpace – A Blogging Platform

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Project Overview

BlogSpace is a database-driven blogging platform designed to manage users, blogs, publications, and their interactions in a structured and efficient manner. The system is built around a well-normalized relational database, ensuring data consistency, minimal redundancy, and scalability.

The platform supports core functionalities such as user registration, blog creation and publishing, following authors and publications, category-based personalization, and engagement features like likes and comments. It also introduces advanced features such as publications with ownership and membership roles, draft management, and a reporting system for moderation.

From a technical perspective, the project focuses on designing an optimized database schema (in BCNF), implementing relationships using primary and foreign keys, and efficiently querying data using SQL. The backend, developed in Python, interacts with the database to handle all operations, while the frontend provides a clean interface for user interaction.

Project Setup and Execution

1. System Prerequisites

- **Python 3.8+** (Required for Django)
- **PostgreSQL** (Primary database for custom SQL and application data)
- **Git** (Optional, for cloning the repository)

2. Database Configuration

The platform uses SQLite for basic Django sessions, but relies on a custom PostgreSQL connection (`db.py`) for core application data.

1. **Create the Database:** Open PostgreSQL (e.g., via pgAdmin) and create a database named `blogging_database`.
2. **Configure Credentials:** By default, `db.py` connects to localhost using the username `postgres` and password `vignasri`. Modify `db.py` if your local credentials differ.
3. **Seed the Data:** Execute the provided SQL scripts (`sql commands.txt` and `sql insert commands.txt`) in your PostgreSQL query tool to build the tables and inject mock data (users, blogs, comments, etc.).

3. Installation & Execution

Open your terminal in the project root directory (where manage.py is located) and run the following commands sequentially:

1. Create and activate a virtual environment

```
python -m venv venv
```

```
source venv/bin/activate # On Windows use: venv\Scripts\activate
```

2. Install required dependencies

```
pip install django psycpg2
```

3. Apply Django's internal setup migrations

```
python manage.py makemigrations
```

```
python manage.py migrate
```

4. Start the development server

```
python manage.py runserver
```

4. Access & Testing

1. Open a web browser and navigate to <http://127.0.0.1:8000/>.
2. **Test Role-Based Access** using the pre-seeded PostgreSQL accounts:
 - **Admin:** Log in as alexandervanguard@platform.com.
 - **Author:** Log in using an author profile (e.g., Ada Lovelace) or create an account and become an author.
 - **Reader:** Log in using a profile (e.g., ithorne@outlook.com) or create an account and start blogging.

Technology Stack Used

1. Backend Technologies

- Programming Language: Python 3.x
- Web Framework: Django

- Database Driver: psycopg2 (Python-PostgreSQL Database Adapter)

2. Database Architecture

- Primary Database: PostgreSQL - Handles core application data, complex relational logic, custom SQL operations, and transaction management.
- Secondary Database: SQLite (dummy.sqlite3) - Manages Django's built-in session state and basic authentication routing.

3. Frontend Technologies

- Markup Language: HTML5
- Styling: CSS3
- Scripting: JavaScript (ES6)
- Rendering Engine: Django Templating Engine (for dynamic data injection and conditional UI rendering).

4. Development & Deployment Tools

- Version Control: Git & GitHub
- Environment Management: Python venv (Virtual Environments)
- Database Management Tools: pgAdmin (utilized for executing complex SQL schema creation and mock data seeding scripts).

Key Platform Functionalities

1. User Account & Profile Management

- **Custom Profiles:** Users can create personalized accounts, select custom avatars, and choose their favorite reading topics to tailor their experience.
- **Account Security:** Includes secure login, password recovery via personalized security questions, and complete account deletion options.

2. Content Creation & Publishing (For Authors)

- **Drafting System:** Authors can write articles using a dedicated editor, save them as private drafts, and publish them when ready.
- **Premium Content Locking:** Authors can choose to make their articles free for everyone or lock them behind a "Premium Only" paywall.
- **Automatic Tagging:** The platform automatically extracts keywords from article titles to help index and organize content.

3. Content Discovery & Reader Engagement

- **Personalized Feeds:** The homepage automatically recommends articles based on the specific topics a user selected during registration.
- **Interactive Reading:** Users can like, comment on, and discuss articles with other readers and authors.
- **Personal Library:** Readers can bookmark their favorite articles to save them for later reading in a dedicated "Library" tab.
- **Following System:** Users can follow their favorite authors or publications to curate their content.
- **Advanced Search:** A robust search bar allows users to find specific articles, topics, or authors quickly.

4. Monetization & Creator Support

- **Premium Subscriptions:** Readers can purchase monthly or yearly subscriptions to unlock access to exclusive, high-quality premium articles.
- **Direct Tipping (Donations):** A built-in financial support system allows readers to send monetary donations directly to the authors of articles they enjoyed.
- **Author Earnings Dashboard:** Creators have a dedicated ledger to track the donations they receive from their audience.

5. Collaborative Publications (Digital Magazines)

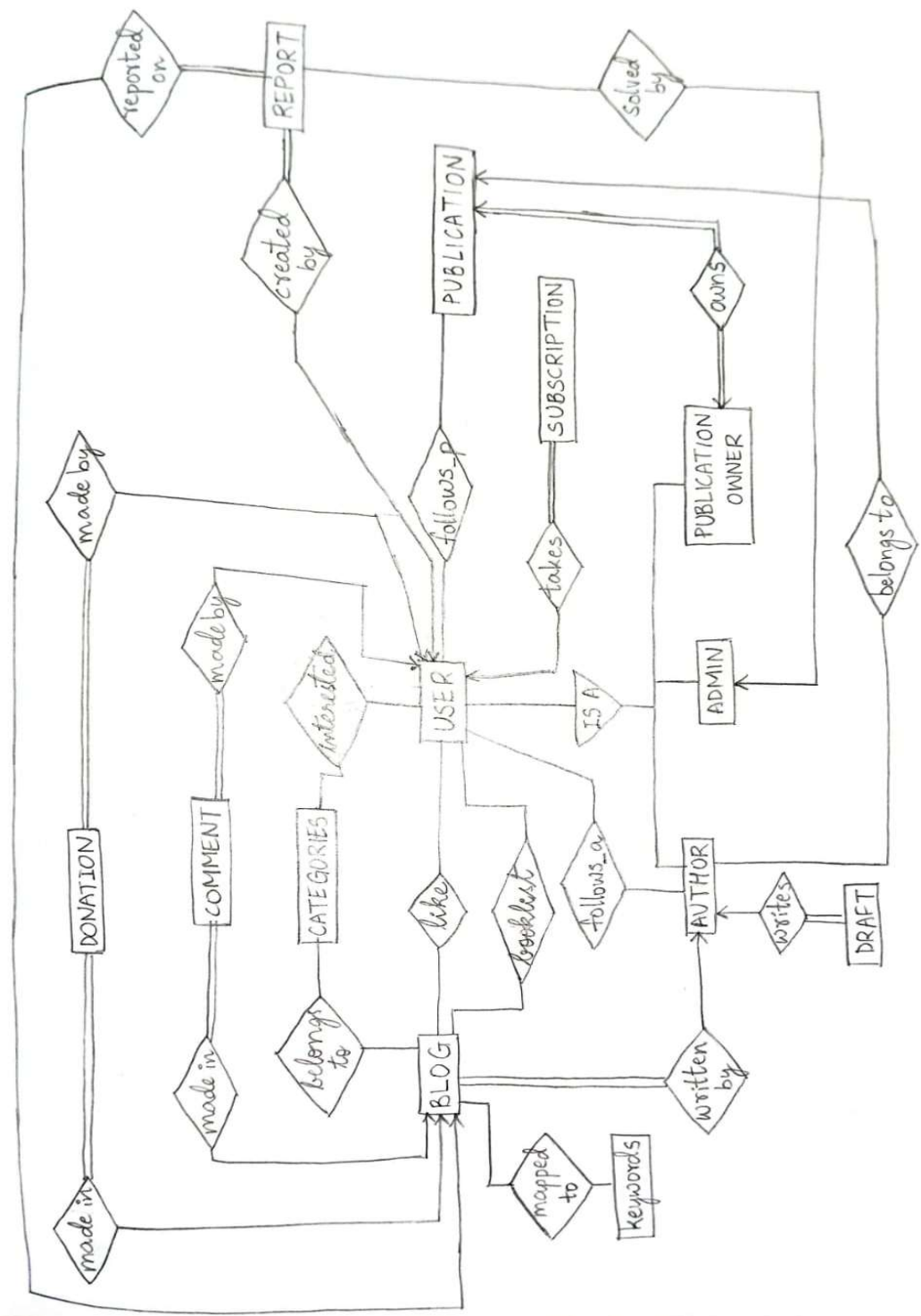
- **Create and Manage:** Authors can launch their own "Publications" (collaborative magazines) and act as the owner.
- **Join Requests:** Other writers can apply to join a publication, and the owner can accept or reject these applications from a dedicated dashboard.
- **Collective Feeds:** Readers can view dedicated pages that aggregate all the articles written by members of a specific publication.

6. Community Moderation & Administration

- **Content Reporting:** Any user can flag an article for violating platform rules (e.g., spam, plagiarism, unmarked AI content) providing details for the report.
- **Admin Dashboard:** Platform administrators have a secure dashboard to review flagged content, monitor platform metrics, and take action (like removing blogs or banning users) to maintain community standards.

Database Design and Architecture

E-R Diagram



Database Schema and Data Dictionary -

Tables created and constraints used -

1. **users** - Central entity storing core account information for anyone registered on the platform.
 - user_id (Integer / Primary Key) - Auto-incremented unique identifier for the user.
 - first_name (Varchar) - The user's primary given name.
 - last_name (Varchar) - The user's surname or family name.
 - email_id (Varchar) - The user's unique email address used for login and communication.
 - password (Varchar) - The securely hashed password for account authentication.
 - premium (Smallint) - A numerical flag (0 or 1) indicating if the user has an active premium subscription.
 - avatar_id (Integer) - A numeric reference to the profile picture selected by the user.
 - security_question (Varchar) - The custom question selected by the user for password recovery.
 - security_answer (Varchar) - The hashed answer to the user's security question.
2. **admins** - Represents platform administrators who manage the site.
 - admin_id (Integer / Primary & Foreign Key) - Matches the user_id to uniquely identify the admin.
 - role (Varchar) - Defines the specific administrative permissions role (e.g., 'super_admin').
 - salary (Integer) - The compensation amount assigned to the administrator.
3. **authors** - Represents content creators on the platform.
 - author_id (Integer / Primary & Foreign Key) - Matches the user_id to uniquely identify the author.
 - publi_id (Integer / Foreign Key) - Links the author to a specific publication they contribute to.
4. **subscriptions** - Tracks the premium subscription plans purchased by users to access exclusive content.
 - sub_id (Integer / Primary Key) - Auto-incremented unique identifier for the subscription record.
 - user_id (Integer / Foreign Key) - Identifies the specific user who purchased the plan.

- `plan_type` (Varchar) - Specifies whether the chosen plan duration is 'monthly' or 'yearly'.
 - `start_date` (Date) - The date the premium subscription was activated.
 - `end_date` (Date) - The calculated expiration date of the subscription.
 - `amount` (Integer) - The total monetary cost paid for the subscription plan.
- 5. blogs** - The primary content table that stores the articles, essays, or posts created by authors.
- `blog_id` (Integer / Primary Key) - Auto-incremented unique identifier for the blog post.
 - `content` (Text) - The full HTML/rich-text body of the blog article.
 - `author_id` (Integer / Foreign Key) - Identifies the specific author who wrote the blog.
 - `update_datetime` (Timestamp) - The exact date and time the post was last modified or saved.
 - `status` (Smallint) - Indicates if the blog is an unpublished draft (0) or publicly published (1).
 - `title` (Text) - The headline or main title of the blog post.
 - `is_premium` (Boolean) - A true/false flag locking the post behind the premium subscriber paywall.
- 6. categories** - Maintains a predefined list of topics or genres that blogs can be grouped under.
- `category_id` (Integer / Primary Key) - Auto-incremented unique identifier for the category.
 - `category_name` (Varchar) - The unique, descriptive name of the topic (e.g., 'Technology').
- 7. publications** - Represents a collective or digital magazine managed by an author where multiple authors can contribute.
- `publi_id` (Integer / Primary Key) - Auto-incremented unique identifier for the publication.
 - `publi_name` (Varchar) - The unique name of the digital magazine or collective.
 - `date_of_joining` (Date) - The official creation date of the publication.
 - `owner_id` (Integer / Foreign Key) - Identifies the lead author who manages and owns the publication.
- 8. blog_categories** - A junction table mapping blogs to their associated categories.

- **blog_id** (Integer / Composite Primary & Foreign Key) - Identifies the specific blog post being categorized.
 - **category_id** (Integer / Composite Primary & Foreign Key) - Identifies the specific category assigned to the blog.
- 9. blog_keywords** - Stores specific tags or keywords associated with individual blogs to aid in searchability.
- **blog_id** (Integer / Composite Primary & Foreign Key) - Identifies the specific blog post.
 - **keyword** (Text / Composite Primary Key) - A single descriptive tag extracted from the blog's title.
- 10. comments** - Stores the text-based responses or discussions left by users on specific blogs.
- **comment_id** (Integer / Primary Key) - Auto-incremented unique identifier for the comment.
 - **description** (Text) - The actual text content of the user's comment.
 - **blog_id** (Integer / Foreign Key) - Identifies the specific blog where the comment was posted.
 - **user_id** (Integer / Foreign Key) - Identifies the specific user who wrote the comment.
 - **creation_datetime** (Timestamp) - The exact date and time the comment was submitted to the platform.
- 11. likes** - Tracks which users have "liked" or upvoted specific blogs.
- **user_id** (Integer / Composite Primary & Foreign Key) - Identifies the user who liked the post.
 - **blog_id** (Integer / Composite Primary & Foreign Key) - Identifies the specific blog post that was liked.
- 12. booklist** - Acts as a "save for later" or bookmarking feature, linking users to the blogs they want to keep in their personal library.
- **user_id** (Integer / Composite Primary & Foreign Key) - Identifies the user saving the post.
 - **blog_id** (Integer / Composite Primary & Foreign Key) - Identifies the specific blog post being bookmarked.
- 13. donations** - Records financial transactions where a reader tips or donates money to an author for a specific blog.
- **donation_id** (Integer / Primary Key) - Auto-incremented unique identifier for the donation transaction.

- `user_id` (Integer / Foreign Key) - Identifies the reader who is making the donation.
- `author_id` (Integer / Foreign Key) - Identifies the specific author receiving the funds.
- `blog_id` (Integer / Foreign Key) - Identifies the specific post that inspired the user to donate.
- `amount` (Integer) - The monetary value of the donation.
- `donated_at` (Timestamp) - The exact date and time the financial transaction occurred.

14. reports - Supports platform moderation by allowing users to flag blogs for rule violations (e.g., spam, plagiarism).

- `report_id` (Integer / Primary Key) - Auto-incremented unique identifier for the moderation ticket.
- `user_id` (Integer / Foreign Key) - Identifies the specific user who filed the report.
- `blog_id` (Integer / Foreign Key) - Identifies the specific blog being flagged.
- `reported_at` (Timestamp) - The exact date and time the report was submitted.
- `type_of_report` (Varchar) - The general category of the violation selected from the UI (e.g., 'Spam' or 'Plagiarism').
- `description` (Text) - The detailed reason or explanation provided by the user in the text box.
- `status` (Varchar) - Indicates if the report is 'new' (pending admin review) or 'solved'.

15. author_followers - Tracks users subscribing to or following specific authors.

- `author_id` (Integer / Composite Primary & Foreign Key) - Identifies the author being followed.
- `user_id` (Integer / Composite Primary & Foreign Key) - Identifies the specific user who is following the author.

16. publication_followers - Tracks users subscribing to or following specific publications.

- `publi_id` (Integer / Composite Primary & Foreign Key) - Identifies the specific publication being followed.
- `user_id` (Integer / Composite Primary & Foreign Key) - Identifies the specific user who is following the publication.

17. interested_categories - Maps a user's personal preferences to specific topics to help tailor their content feed.

- `user_id` (Integer / Composite Primary & Foreign Key) - Identifies the specific user.

- `category_id` (Integer / Composite Primary & Foreign Key) - Identifies the category they have marked as an interest.
- 18. publication_join_requests** - Manages the workflow of users requesting to become contributing authors for a specific publication.
 - `request_id` (Integer / Primary Key) - Auto-incremented unique identifier for the join request.
 - `user_id` (Integer / Foreign Key) - Identifies the specific user applying to join the publication.
 - `publi_id` (Integer / Foreign Key) - Identifies the specific publication they wish to join.
 - `status` (Varchar) - The current state of the application, restricted to 'pending', 'accepted', or 'rejected'.

Normalization of database -

The entire database schema is in **Boyce-Codd Normal Form (BCNF)**. This is because, across all tables, every determinant is a candidate key. There are no partial dependencies (satisfying 2NF), no transitive dependencies (satisfying 3NF), and no overlapping candidate keys that would violate BCNF. Hence data redundancy is minimized, and update anomalies are prevented.

Backend Architecture and Core Modules

These are the core functions written to achieve the backend working of the platform.

1. Authentication & User Management

- **register**: Handles user creation by hashing passwords (`make_password`), executing INSERT queries for user data and topic preferences, and initializing session variables.
- **login_view**: Validates credentials using `check_password` against database hashes, queries for admin privileges, and establishes session state upon success.
- **logout_view**: Terminates the user session using Django's `session.flush()` and redirects to the login route.
- **profile**: Aggregates user metadata, followed entities, and platform roles (Author/Owner) using complex SQL JOINS across multiple relational tables.
- **edit_profile**: Handles POST requests to update user attributes (name, avatar, hashed password) and replaces existing categorical preferences via DELETE and INSERT statements.
- **update_avatar / update_categories**: Dedicated helper views that execute targeted UPDATE or INSERT SQL commands to instantly change visual or categorical preferences.

- **forgot_password:** A two-step validation process that queries the database for a user's security question, validates the hashed answer, and updates the database with a new hashed password.
- **delete_account:** Executes a massive manual cascading delete operation via sequential SQL DELETE commands, removing all user dependencies, interactions, publications, and blogs before deleting the core user row.

2. Core Content & Blog Management

- **write:** Serves the text editor UI; on form submission, it handles SQL INSERT commands to save blogs as either drafts (status=0) or published (status=1) and extracts/stores keywords.
- **extract_keywords:** A Python helper function that normalizes blog titles (lowercasing, removing punctuation) and filters out common English stop-words to generate search tags.
- **edit_draft / delete_draft:** Executes SQL UPDATE to modify or publish existing drafts, or DELETE to permanently erase them from the blogs table.
- **delete_blog:** Verifies user ownership before executing manual cascade deletes of all associated engagements (comments, likes, donations) and finally removing the blog record.
- **stories:** Queries the database to separate and display a user's content into two distinct lists: unpublished drafts and published articles.
- **blog_detail:** Fetches comprehensive blog data (comments, likes, categories) and conditionally renders content by verifying the user's premium subscription status or authorship.

3. Home Feed & Search Engine

- **home (Recommendation Engine):** Generates a personalized feed using a SQL LEFT JOIN algorithm that matches a blog's categories against the user's interested_categories, sorting results by highest match score and recency.
- **author_blogs:** Similar to the home feed, but filters the query to only show published articles from a specific author, while still applying the recommendation matching algorithm.
- **search:** Implements advanced text matching utilizing PostgreSQL's pg_trgm extension and word_similarity(), comparing the query against a dynamically concatenated string of titles, authors, categories, and keywords.

4. User Engagement (Interactions)

- **like_blog / bookmark_blog:** Toggle functions that query junction tables (likes, booklist); if a record exists, it executes a DELETE, otherwise it performs an INSERT.
- **follow_author / follow_publication:** Implements toggle logic similar to likes/bookmarks, modifying the author_followers and publication_followers relationship tables.
- **add_comment:** Captures form input and inserts a new row into the comments table, attaching the current timestamp via SQL's NOW().

- **library:** Executes parallel queries against the booklist and likes tables, joining them with blog data to render the user's saved and liked content feeds.

5. Publications Management

- **create_publication:** Verifies author status, inserts a new collective entity into the publications table, and updates the creator's publi_id to establish ownership.
- **publication_detail:** Aggregates and renders a publication's specific data, including a list of associated authors and their published blogs, utilizing relational JOINS.
- **owner_dashboard:** A specialized view for publication owners that calculates metrics (follower count) and lists active members, published content, and pending join requests.
- **join_request / handle_request:** Manages the application workflow by inserting pending requests into the database, which the owner can later UPDATE to 'accepted' (linking the author) or 'rejected'.
- **transfer_ownership / leave_publication / delete_publication:** Administrative functions for owners/authors that execute specific UPDATE commands to reassign foreign keys or DELETE commands to dissolve the entity.

6. Monetization: Premium & Donations

- **subscribe / subscriptions:** Queries for active premium plans based on end_date. On purchase, utilizes Python's datetime module to calculate expiration dates and inserts a new subscription tier.
- **donate:** Captures financial input and inserts a transactional record linking the donor (user_id), the recipient (author_id), and the content (blog_id), explicitly blocking self-donations.
- **author_donations:** Queries the donations table, joining it with user and blog data, to generate a chronological ledger of financial support received by an author.

7. Moderation & Administration

- **report_blog:** Allows users to flag content by inserting a row into the reports table with a categorized reason and a default status of 'new'.
- **admin_dashboard:** Restricted view that queries the database for unresolved reports alongside high-level platform metrics (latest users, blogs, and comments).
- **admin_action:** A multi-purpose routing function that executes irreversible database commands (e.g., deleting flagged users, taking down blogs, or marking reports as 'solved') based on URL parameters.

Frontend and User Interface

- **Architecture:** Server-Side Rendered (SSR) utilizing the **Django Templating Engine**.
- **Core Technologies:** HTML5, CSS3 (featuring custom glassmorphism and mesh gradients), and vanilla JavaScript (ES6).
- **Dynamic Data Injection:** Utilizes Django context variables and template tags (e.g., `{{ variable }}`, `{% for %}`) to dynamically render database records (blogs, metrics, profiles) directly into the DOM.
- **Role-Based Conditional UI:** Employs logical template tags (`{% if %}`) to enforce access control at the UI level. This includes rendering distinct dashboards for Admins/Authors and displaying "Premium Locked" paywalls based on the active user's session state.
- **Interaction Handling:** Relies on synchronous HTML form submissions (POST) and URL routing mapped to backend Django views to handle state changes (e.g., publishing drafts, toggling likes, or submitting payments) without complex client-side state management.

Conclusion and Future Scope

Conclusion

The development of the BlogSpace platform successfully demonstrates the implementation of a robust, full-stack web application designed to foster high-quality digital discourse. By leveraging the Django web framework in tandem with a highly structured PostgreSQL database, the project achieves a seamless balance between complex backend logic and a dynamic, user-friendly frontend interface.

Key technical milestones achieved during this project include:

- **Relational Database Integrity:** The design and implementation of a comprehensive database schema that achieves Boyce-Codd Normal Form (BCNF) across its core tables, ensuring data consistency while utilizing strategic denormalization to optimize real-world read performance.
- **Role-Based Architecture:** The successful deployment of a multi-tiered access control system that provides distinct, secure environments for Readers, Authors, Publication Owners, and Platform Administrators.
- **Monetization Ecosystem:** The integration of dual revenue streams—platform-level Premium Subscriptions and peer-to-peer Author Donations—backed by secure relational transaction mapping.
- **Community Moderation:** The creation of a self-regulating ecosystem where users can flag content, directly feeding into an actionable Admin Dashboard.

Future Expansions

The following features represent the roadmap for future iterations and platform enhancements:

1. Advanced Author Analytics Currently, authors can track direct financial donations. Future expansions will introduce a comprehensive analytics dashboard utilizing data visualization libraries (such as D3.js or Chart.js). This will allow authors to track granular metrics including "Average Read Time," "Traffic Sources," and "Geographic Demographics" of their readership.

2. Real-Time Notification System To increase user retention, the platform will implement Django Channels and WebSockets to push real-time notifications. Users will receive instant alerts when an author they follow publishes a new essay, when their comment receives a reply, or when a publication join request is accepted.

3. AI-Assisted Moderation and Analytics To ease the burden on platform Administrators, machine learning APIs could be integrated into the reporting pipeline. This would automatically scan newly published blogs for heavily plagiarized text or entirely unmarked AI-generated content, automatically flagging suspicious articles for Admin review before users even need to report them.

4. Cloud Media Storage Integration As the platform scales, handling media files locally will become inefficient. Future versions will migrate static assets and user-uploaded content (such as inline blog images, publication banners, and high-resolution avatars) to cloud storage solutions like AWS S3 or Google Cloud Storage, served via a Content Delivery Network (CDN) for faster global load times.

5. Audio "Listen" Feature Given the platform's focus on long-form, deep-dive journalism, implementing a Text-to-Speech (TTS) engine would allow readers to listen to articles as podcasts.

Contributions

- V. Varshitha: Implemented the frontend using HTML, CSS, and JavaScript to create a responsive design.
- B.Sri Hasree: Implemented the backend using Python and Django, and connected it to the database.
- G. Naga Manikanta: Created the schema design in PostgreSQL, developed the required tables, and inserted data into them.

Github Link: <https://github.com/Hasree13/Blogging-platform>

Screenshots of Blogging platform-

